Two-Sample Mean Inference

Estimating a difference in means

Download the section 14 .Rmd handout to `STAT240/lecture/sect14-two-means`.

Material in this section is covered by Chapter 13 on the notes website.

Download the file `cat_breeds_clean.txt` to `STAT240/data`.

Continue working with the cat sleep data.

What is the change in the average sleep time for fixed male ragdoll cats versus intact male ragdoll cats?

Inference for $\mu_1 - \mu_2$:

- Point estimate: $\bar{X}_1 - \bar{X}_2$
- Standard error:

$$\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

- Model: T distribution

Assume the groups have two different populations:

$$X_{1,i} \sim D_1(\mu_1,\ \sigma_1)$$
$$X_{2,i} \sim D_2(\mu_2,\ \sigma_2)$$

and populations 1 and 2 are independent.

We allow the distributions to differ completely. All that matters is $\bar{X}_1$ and $\bar{X}_2$ are approximately normal.

$$\text{point estimate} \pm \text{critical value} \times \text{standard error}$$

- For $\mu_1 - \mu_2$, the point estimate is $\bar{x}_1 - \bar{x}_2$
- The standard error combines σ_1 and σ_2

The standard error is

$$\sqrt{se_1^2 + se_2^2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

This gives sampling distribution

$$\bar{X}_1 - \bar{X}_2 \quad \dot{\sim} \quad N\left(\mu_1 - \mu_2, \ \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right)$$

This suggests a normal critical value. However, we need to estimate the se.

$$\hat{se}(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Again, we use a T distribution for our CI and hypothesis test.

There are two different types of two-sample T test:

- Equal variances T (often more powerful)
- Unequal variances T (more general)

We can assume equal variances when the sample SDs are within a factor of 2.

$$0.5 < \frac{s_1}{s_2} < 2$$

When we estimate $\sigma_1 = \sigma_2 = \sigma$, we can simplify the standard error.

$$se = \sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Estimate σ with the **pooled sd** s_p.

The pooled sd is a weighted average of the groups.

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

The weights are $n_1 - 1$ and $n_2 - 1$.

In the cats example, the pooled sd is about 2.477.

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

- Is s_p more similar to s_1 or s_2?
- When is s_p an exact average of s_1 and s_2?

★ Why do we square the terms, then take an average, then square root? Why not just average s_1 and s_2?

Now, we can estimate the standard error.

$$\hat{se} = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Lastly, we use a T with $n_1 + n_2 - 2$ degrees of freedom for the critical value.

We are 90% confident that the difference in sleep time is within (-0.91, 0.35). More generally, we have

$$\bar{X}_1 - \bar{X}_2 \pm t_{\alpha/2, n_1+n_2-2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

We can also perform a hypothesis test for the difference in means. Let's do a 10% level test of the difference in sleep times:

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{versus} \quad H_A : \mu_1 - \mu_2 \neq 0$$

The test statistic (assuming equal variances) is

$$T = \frac{\bar{X}_1 - \bar{X}_2 - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

and follows a T with $n_1 + n_2 - 2$ degrees of freedom if H_0 is true.

In general, we can test for any mean difference μ_0:

$$H_0 : \mu_1 - \mu_2 = \mu_0 \quad \text{versus} \quad H_A : \mu_1 - \mu_2 \neq \mu_0$$

with test statistic

$$T = \frac{\bar{X}_1 - \bar{X}_2 - \mu_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

What if we aren't willing to make the equal variances assumption? We can still perform a similar T analysis using the more general standard error:

$$\hat{se}(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Let's compare intact male cats and kittens.

The T is approximate, and the df is messy.

$$w = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}$$

This is the **Welch** procedure. The df are "penalized" according to s_1 and s_2.

The formula for the CI is

$$\bar{X}_1 - \bar{X}_2 \pm t_{\alpha/2,w} \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

We are 95% confident that the true difference is within (-3.01, -0.54). `t.test()` confirms this.

Are adult cats more sleepy than kittens?

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{versus} \quad H_A : \mu_1 - \mu_2 > 0$$

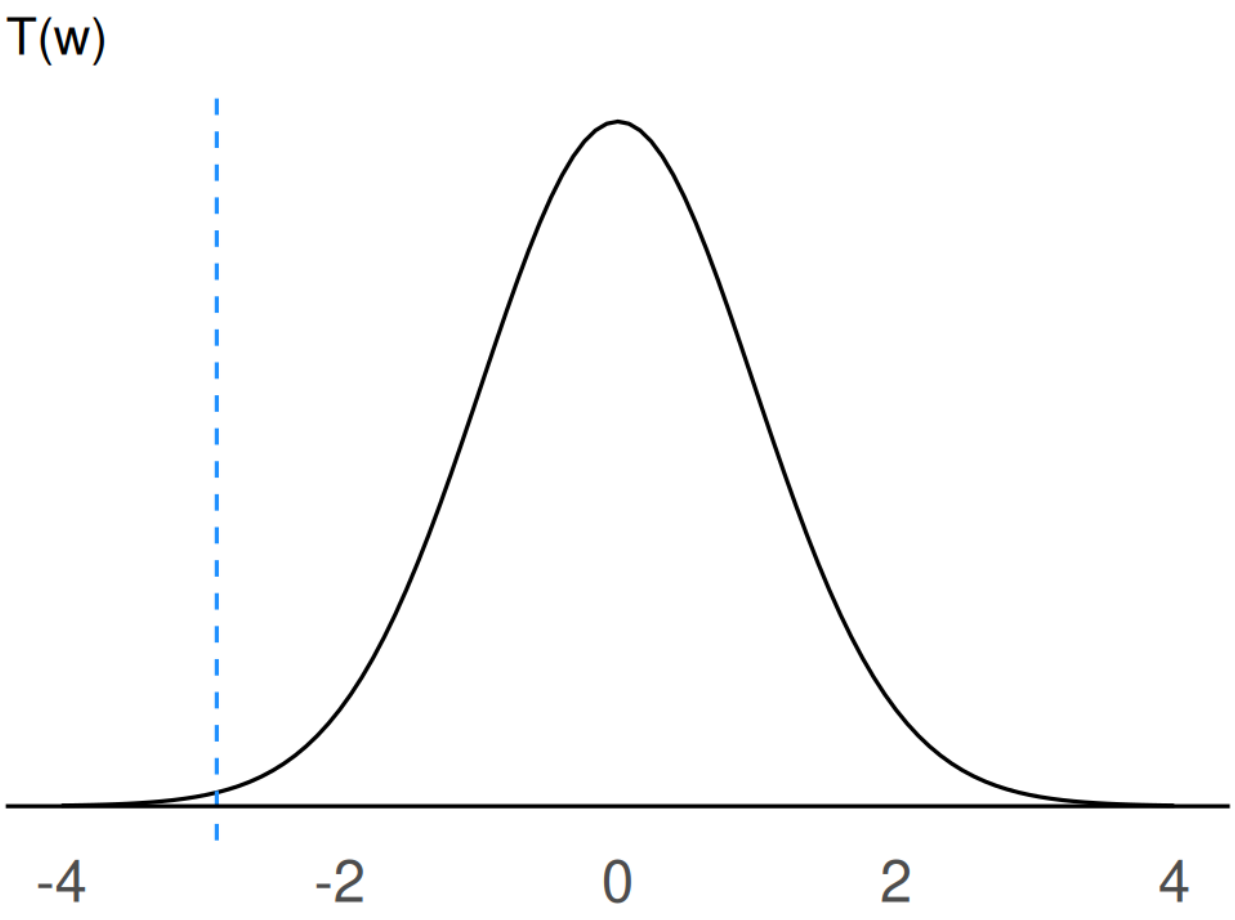
Let's complete this test with $\alpha = 0.05$.

Our observed test statistic is

$$T = \frac{\bar{X}_1 - \bar{X}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

If H_0 is true, T has a T distribution with w degrees of freedom.

- Complete the test of the one-sided hypotheses.


T(w)
-4
-2
0
2
4

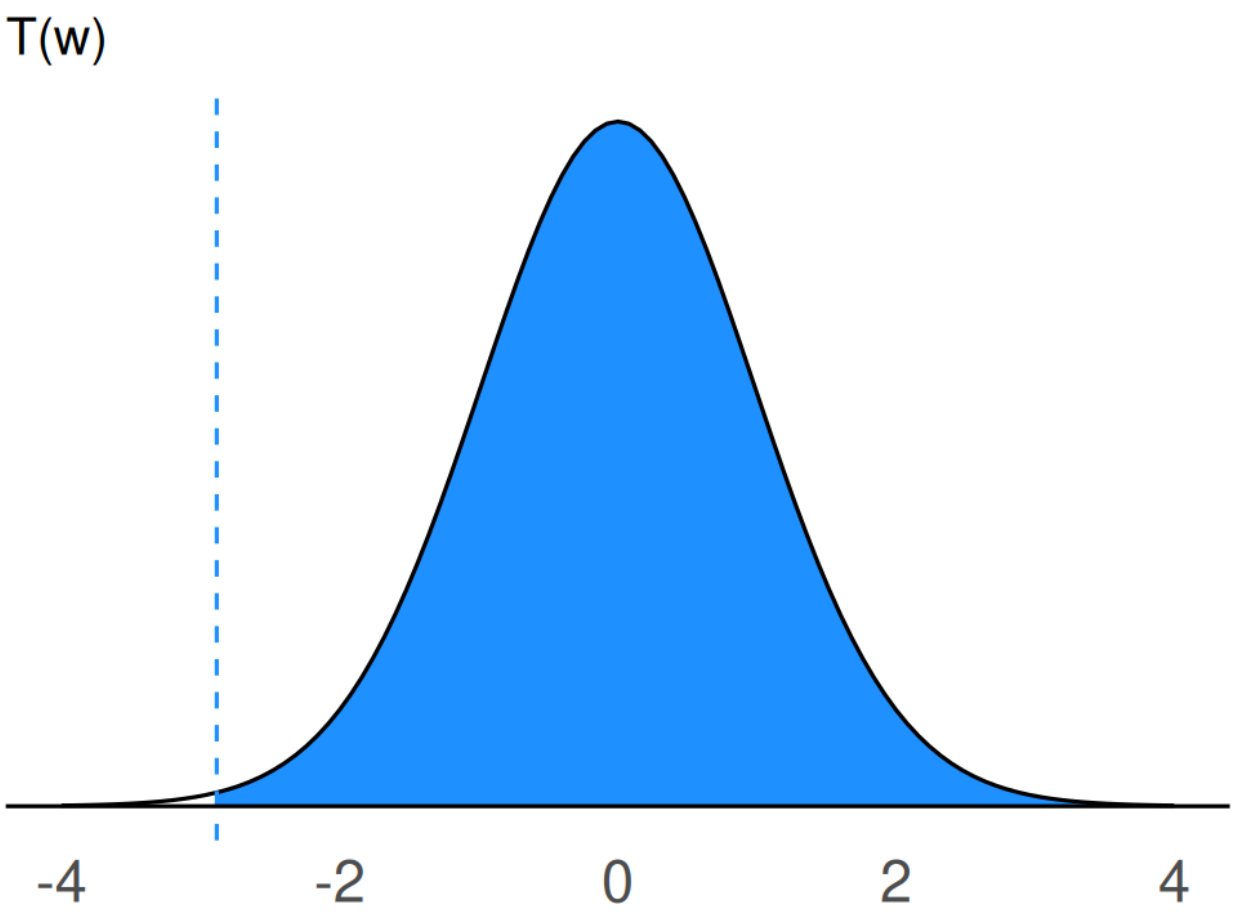
For hypotheses

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{versus} \quad H_A : \mu_1 - \mu_2 > 0$$

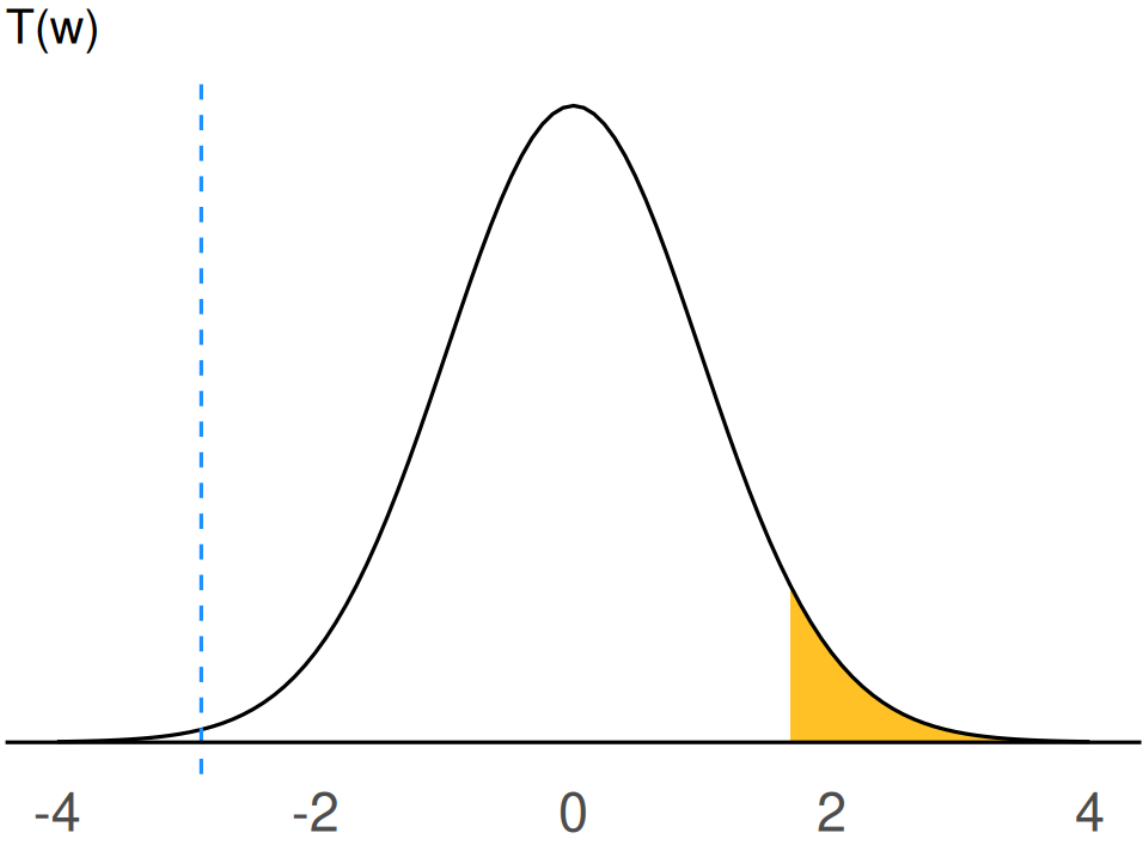
we have p-value

$$P(T_w > 0.316) = 0.997$$

and fail to reject H_0.


T(w)
-4
-2
0
2
4

$\alpha = 0.05$ rejection region:



P-value calculation:

- If the alternative is H_A : diff $< \mu_0$, the p-value is the area *below* the test statistic.
- If the alternative is H_A : diff $> \mu_0$, the p-value is the area *above* the test statistic.
- If the alternative is H_A : diff $\neq \mu_0$, the p-value is $2\times$ the area *outside* of the test statistic.

What if we assumed equal variances? In general:

- If the variances are truly equal, the Welch T test is a bit less powerful, but is still accurate.
- If the variances are truly different, the equal variances test can make incorrect conclusions.

It is safer to assume unequal variances.

Sometimes, two samples of data are **paired**.

Do children grow taller from age 13 to 14? It doesn't make sense to take independent samples of 13 and 14 year olds.

Instead, measure the same individuals at 13 and 14.

Five individuals had their heights measured at age 13 and 14.

Height at 13	44.1	59.0	65.9	58.7	49.3
Height at 14	46.3	60.5	68.2	59.4	50.6

The two samples are *not independent*, so we can't use one of the previous methods.

Instead, study the *differences*.

Height at 13	44.1	59.0	65.9	58.7	49.3
Height at 14	46.3	60.5	68.2	59.4	50.6
Difference	2.2	1.5	2.3	0.7	1.3

We are back in a one-sample setting! Parameter:

$$\mu_{diff} \quad \text{instead of} \quad \mu_1 - \mu_2$$

Assume the differences are independent draws from a population:

$$diff_i \sim D(\mu, \sigma)$$

We then have hypotheses

$$H_0 : \mu_{diff} = 0 \quad \text{versus} \quad H_A : \mu_{diff} > 0$$

We use the usual one-sample T test statistic

$$T = \frac{\bar{diff} - \mu}{S/\sqrt{n}}$$

which follows a T with $n - 1$ df if H_0 is true.

Let's complete the test with $\alpha = 0.05$.

We have a large positive test statistic and a small p-value, so we reject H_0.

This is confirmed in `t.test()`.

If we had done a two independent sample test, we get the opposite result!